

PRACTICUM WORKSHEET: THE AMDAHL REALITY GAP

Student Full Name: Abisheva Assem

Student ID Number: 230103284

Institutional Email / Git URL: 230103284@gmail.com / [asemabiseva/314_17-09_230103284](https://github.com/asemabiseva/314_17-09_230103284)

Practicum Date & Time (Slot): 24.09 02-N Thursday 11:30-13:30

Laptop Model & OS (Linux/macOS/Win): Huawei matebook d15 / Win

CPU Model & Generation: AMD Ryzen 5 5500U with Radeon Graphics

Physical Cores (P/E breakdown if hybrid): 6

Logical SMT Threads (omp_get_max_threads): 12

Target Workload Size N (Derived via Formula): 10 284 000

L1 / L2 Cache Line Size (Typically 64 bytes): _____

1.2 results

Threads: 1 | Max stopping time: 685 | Time: 6342,5224 ms

Threads: 2 | Max stopping time: 685 | Time: 3500,8153 ms

Threads: 4 | Max stopping time: 685 | Time: 1863,7612 ms

Threads: 8 | Max stopping time: 685 | Time: 1021,3514 ms

Threads: 16 | Max stopping time: 685 | Time: 835,8539 ms

2.1 Results

N: 10284000

Max stopping time: 685

Checksum: 599305808

Time: 5111,2901 ms

2.2 Results

N = 10284000

Run 1 (Warmup)

Max stopping time: 685 | Checksum: 599305808

Run 1 discarded.

Run 2:

Max stopping time: 685 | Checksum: 599305808

Run 2 time: 5260,2106 ms

Run 3:

Max stopping time: 685 | Checksum: 599305808

Run 3 time: 5069,0588 ms

$T_{seq} = (\text{Run 2} + \text{Run 3}) / 2 = 5164,6347 \text{ ms}$

3 task

N = 10284000

CPU cores: 12

Starting k = 1...

k=1 | Run1=324.4386s | Run2=236.6635s | Run3=252.9658s | Avg=244.8146s | Max=685 |
Checksum=599305808

Starting k = 2...

k=2 | Run1=126.7059s | Run2=115.3446s | Run3=115.7971s | Avg=115.5709s | Max=685 |
Checksum=599305808

Starting k = 4...

k=4 | Run1=71.1725s | Run2=70.1218s | Run3=82.9958s | Avg=76.5588s | Max=685 |
Checksum=599305808

Starting k = 8...

I didn't have enough time to calculate all k's

```
N = 10284000  
CPU cores: 12
```

```
Starting k = 1...
```

```
k=1 | Run1=324.4386s | Run2=236.6635s | Run3=252.9658s | Avg=244.8146s | Max=685 | Checksum=599305808
```

```
Starting k = 2...
```

```
k=2 | Run1=126.7059s | Run2=115.3446s | Run3=115.7971s | Avg=115.5709s | Max=685 | Checksum=599305808
```

```
Starting k = 4...
```

```
k=4 | Run1=71.1725s | Run2=70.1218s | Run3=82.9958s | Avg=76.5588s | Max=685 | Checksum=599305808
```

```
Starting k = 8...
```

```
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